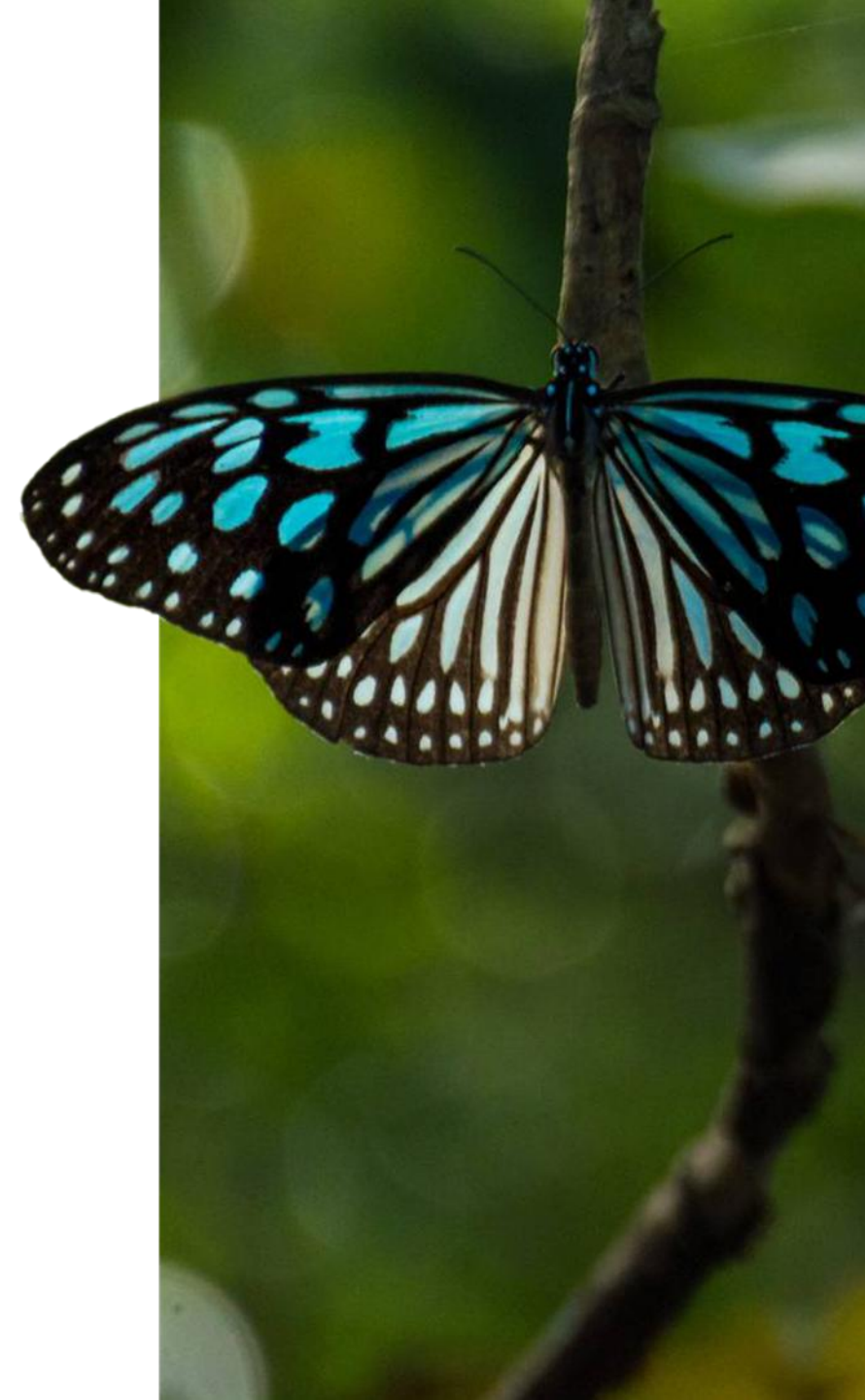


GREEN VALLEY MEDICAL CENTER

Hospital Performance Analysis

Overview of Patient Outcomes, Cost & Operational Performance

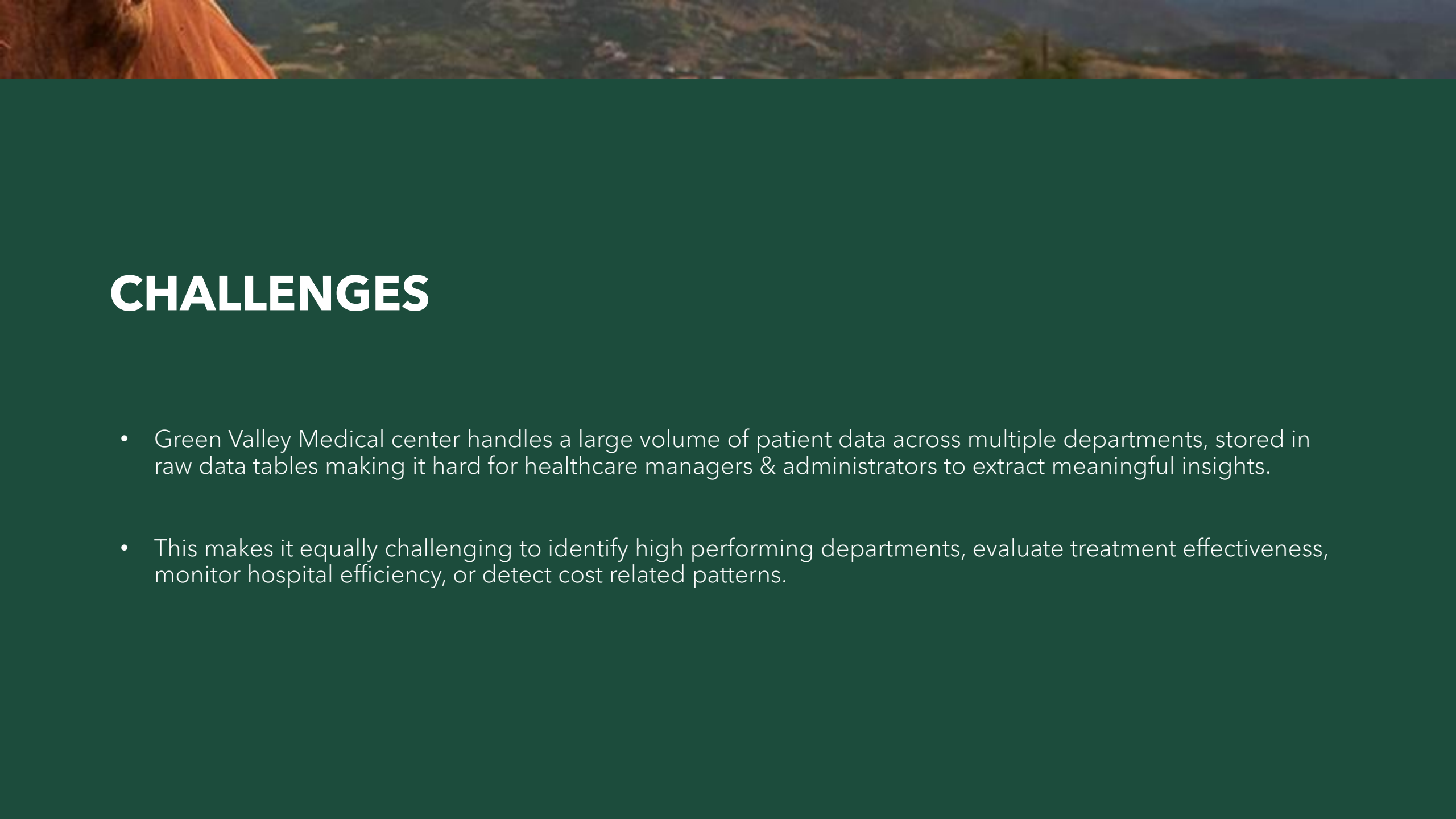


BUSINESS INTRODUCTION

Green Valley Medical Center is a Multi-departmental healthcare facility that provides a wide range of medical services, including cardiology, orthopedics, neurology, pediatrics and general medicine. The hospital is dedicated to delivering quality patient care through effective diagnosis, treatment, and recovery management.

To support better decision-making and improve patient outcomes, the hospital collects and manages data related to patient demographics, treatment types, hospital stays, treatment cost, and recovery performance across its departments.

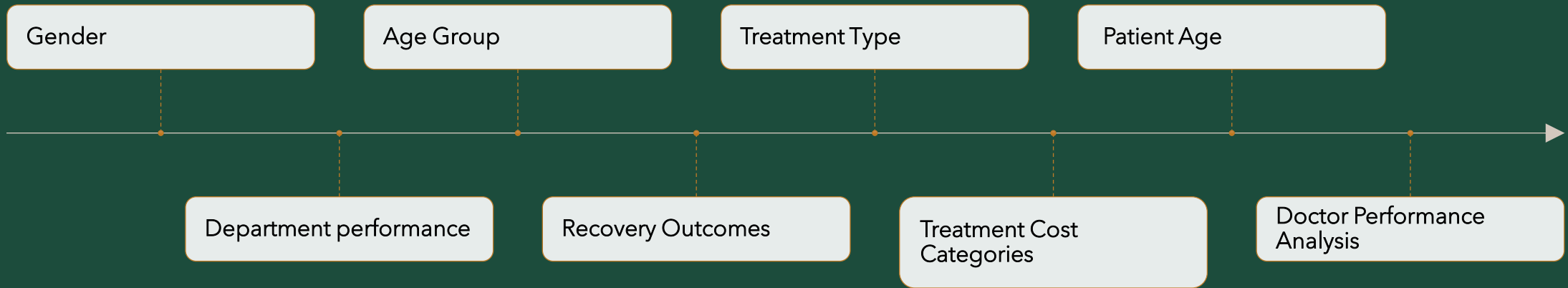
KPI's	Value
Total Patients	400
Avg Recovery Score	75.34
Average Stay	5 Days
Average Cost	£5,051.53
Highest Performing Department	Dermatology
Lowest Performing Department	Cardiology



CHALLENGES

- Green Valley Medical center handles a large volume of patient data across multiple departments, stored in raw data tables making it hard for healthcare managers & administrators to extract meaningful insights.
- This makes it equally challenging to identify high performing departments, evaluate treatment effectiveness, monitor hospital efficiency, or detect cost related patterns.

PRIMARY ANALYSIS AREAS FOR ENHANCING PERFORMANCE REVIEW





TOTAL PATIENTS



400

AVG. TREATMENT COST



£5,051.53

AVG. RECOVERY SCORE



75.34

AVERAGE AGE



45

AVG. HOSPITAL STAY (DAYS)



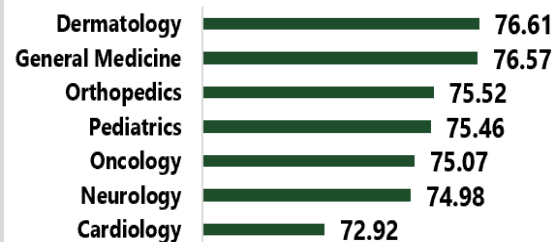
5

BEST PERFORMING DOCTOR

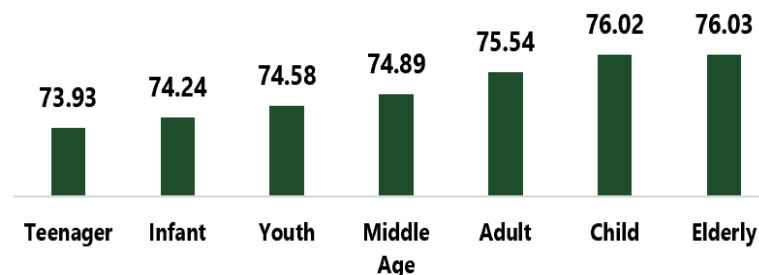


DR. JOHNSON

DEPT. PERFORMANCE BY AVG. RECOVERY SCORE



AVG. RECOVERY SCORE BY AGE GROUP



INSIGHTS



Dermatology has the highest recovery score indicating effective treatment outcomes



Significant gap between top and bottom performing doctors

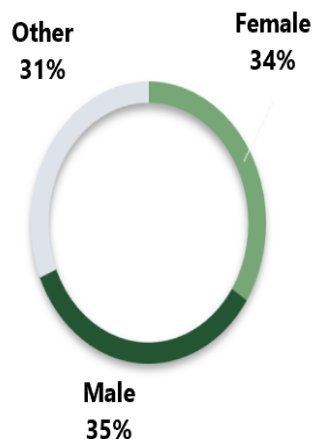


Dr. Smith & Rodriguez underperform in recovery scores, suggesting an opportunity for process improvement.

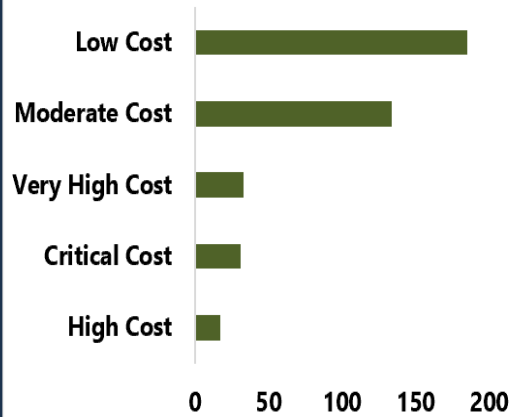


Cost & Category: Low and Moderate -cost patients dominate

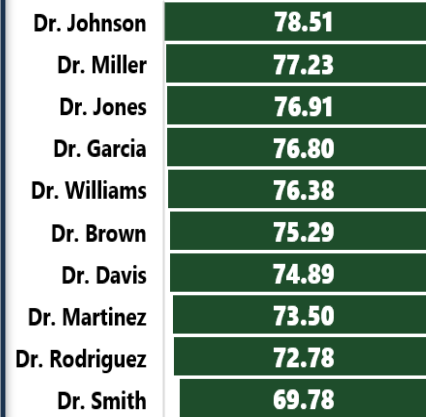
RECOVERY BY GENDER



TREATMENT COST BY CATEGORY



DR. PERFORMANCE BY AVG. RECOVERY SCORE



RECOMMENDATION

Prioritize resources and training for department showing lower recovery stats(Cardiology & Oncology).

Standardise Surgical and Dermatology procedures treatment protocols for optimal patient recovery scores across all departments

Schedule performance reviews for Dr, Rodriguez and Dr. smith, focusing on lower-than-avg. recovery scores.

Focus on cost-optimization while maintaining quality investigate if cardiology low score are due to higher acuity, understaffing or outdated treatment protocols

Department



Cardiology

Dermatol...

General M...

Neurology

Oncology

Orthopedics

Pediatrics

Doctor Name



Dr. Brown

Dr. Davis

Dr. Garcia

Dr. Johnson

Dr. Jones

Dr. Martinez

Dr. Miller

Dr. Rodriguez

Dr. Smith

Dr. Williams

Treatment Type



Medication

Observation

Surgery

Therapy

GREEN VALLEY MEDICAL CENTER ANALYTICS & PERFORMANCE REVIEW

We have identified important key trends that demand our attention.

- Data reveals a stable trend in consumer demand for services and recovery outcomes regarding hospital performance
- Dermatology achieved the highest recovery scores, suggesting that treatment protocols used within the department may offer opportunities for replication across lower-performing departments.
- Cardiology had the lowest average recovery score among all departments, suggesting the need for process review or patient care improvement.
- Most patients fell into the Elderly and Middle Age categories, indicating a higher demand for adult and elderly healthcare services.
- Low-Cost treatments accounted for the highest proportion of cases, showing that most treatments were relatively affordable or low-complexity.
- Moderately Effective treatments were the most common outcome category, while Highly Effective treatments represented a smaller proportion.
- The average hospital stay was approximately 5 days, suggesting that most cases were short-to-medium duration admissions.
- Male patients represented 53% of admissions compared to 47% female admissions, although the gender distribution remained generally balanced.
- A small number of cases were classified under "Other" gender, which may not be statistically significant for broad conclusions.

IMPLICATIONS OF CURRENT STATE OF THINGS

1. Treatment Cost Distribution Insight :

Low-cost treatments accounted for the largest proportion of patient cases.

Financial Implication: This suggests that Green Valley Medical Center relies heavily on high-volume, lower-revenue services. While these services may generate steady cash flow, they may not contribute significantly to overall profitability compared to more specialized treatments.

2. Departmental Cost Performance Insight:

Some departments incur higher treatment costs while not necessarily delivering the highest recovery outcomes.

Financial Implication: Departments with high treatment costs and relatively lower recovery scores may be consuming resources inefficiently. Management should evaluate staffing levels, treatment protocols, equipment utilization, and procurement costs.

3.Length of Stay and Cost Relationship Insight:

The average hospital stay is approximately 5 days.

Financial Implication:

- Longer stays increase:
- Bed occupancy costs
- Staffing expenses
- Medication costs
- Utility and operational expenses



IMPLICATION OF CURRENT STATE OF THINGS CONTINUED

4. Elderly Patient Demographics Insight:

- Most patients fall within Middle-Aged and Elderly categories.
- Financial Implication:
- Older patients often require:
- More complex treatments
- Longer recovery periods
- Additional diagnostic tests
- Increased monitoring

This may increase overall treatment costs and resource consumption.

6. Revenue Opportunity Insight:

The dataset indicates a strong volume of low-cost procedures.

Financial Implication:

While these procedures provide consistent revenue, the hospital may have opportunities to expand higher-value specialist services where clinically appropriate.



RECOMMENDATION

- Investigate the Cardiology department to identify causes of lower recovery performance and introduce targeted quality improvement measures.
- Increase focus on preventive care and elderly patient support since elderly patients represent the largest demographic group.
- Develop strategies to improve treatment effectiveness from “Moderately Effective” to “Highly Effective” outcomes.
- Collect and put together time-based data to ensure monthly or quarterly performance improvements can be monitored.
- Schedule performance reviews for Dr. Rodriguez and Dr. Smith, focusing on lower-than-average recovery scores.
- Include patient satisfaction or readmission feedback to strengthen performance evaluation and healthcare quality assessment.
- Investigate child recovery with focused patient care pathways, streamline low-cost treatment pathways to improve patient flow and operational efficiency.”
- Reducing unnecessary hospital stays can improve bed availability and increase patient turnover.
- Implement discharge planning initiatives and care pathway optimization.
- Develop specialized geriatric care pathways to improve outcomes while controlling costs
- Evaluate demand for Cardiology services, Neurology services, Orthopedic procedures, Preventive health programs to diversify revenue streams.
- Develop specialized geriatric care pathways to improve outcomes while controlling costs.





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